

SEDAR 28: Gulf of Mexico and South Atlantic Cobia and Spanish Mackerel

South Atlantic Cobia:

Data Workshop

Life History

- The LHWG recommends implementation of a tagging study along the entire east coast of Florida and the evaluation of genetic samples from the same to determine more precise stock boundaries.
- Recommend developing a tagging program for inshore and offshore South Atlantic Cobia populations. The goal would be to deploy tags inshore during the spring migration and offshore during the fall and winter to get a clearer picture of fall and spring migrations and to better identify spawning areas and aggregations.
- Explore the feasibility of satellite tags for Cobia movement studies.
- Provide genetic sampling kits to interested groups to better understand the stock division line between the Gulf and Atlantic Cobia stocks. Possible collectors of genetic samples could include Charter operators, fishing clubs and state fisheries personnel.
- Further research is needed on Cobia and Spanish mackerel release mortality.
- To increase the overall amount of data available on Cobia, it is recommended that port samplers do complete workups when sampling, including otolith removal for aging, length, weight, sex, genetic sampling and record a catch location.

Commercial Statistics

Although under the category of research recommendations, this list is not research per se, but rather suggestions to improve data collection. The first three recommendations were modified from the SEDAR17 DW report.

- Need to expand observer coverage.
- Expand TIP sampling to better cover all statistical strata.
- Trade off with lengths versus ages, need for more ages (i.e., hard parts).
- Consider the use of VMS to improve spatial resolution of data.
- During discussions at the data workshop it was noted that the logbook categories for discards (all dead, majority dead, majority alive, all alive) are not useful for informing discard mortality. Consider simplified logbook language in regard to discards (e.g., list them as dead or alive).
- Uniformity between state and federal reporting systems/forms would vastly improve the ease and efficiency of data compilation.
- Establish online reporting and use logbooks as a backup.
- Establish a mechanism for identifying age samples that were collected by length or market categories, so as to better address any potential bias in age compositions.
- Compiling commercial data is surprisingly complex. As this is the 28th SEDAR, one might expect that many of the complications would have been resolved by now through better coordination among NMFS, ACCSP, and the states. Increased attention should be given toward the goal of "one-stop shopping" for commercial data.

Recreational Statistics

- Increase proportion of fish with biological data within MRFSS sampling.
- Continue to develop methods to collect a higher degree of information on released fish (length, condition, etc.) in the recreational fishery.
- Require mandatory reporting for all charter boats state and federal.
- Continue development of electronic mandatory reporting for for-hire sector.
- Continued research efforts to incorporate/require logbook reporting from recreational anglers.
- Establish a review panel to evaluate methods for reconstructing historical landings (SWAS, FWS, etc.).
- Quantify historical fishing photos for use in reconstructing recreational historical landings.
- Narrow down the sampling universe. Identify angler preference and effort. Require a reef fish stamp for anglers targeting reef fish, pelagic stamp for migratory species, and deepwater complex stamp for deep-water species. The program would be similar to the federal duck stamp required of hunters. This would allow the managers to identify what anglers were fishing for.
- Continue and expand fishery dependent at-sea-observer surveys to collect discard information, which would provide for a more accurate index of abundance.

Indices

- Explore SEFIS video data as a potential fishery independent index of abundance for cobia.
- Using simulation analysis, evaluate the utility of including interaction terms in the development of a standardized index and identify the potential effects these interaction terms have on stock assessments.

Assessment Workshop

The assessment panel made the following recommendations.

- Develop a fishery independent sampling program for abundance of cobia and other coastal migratory species. Fishery dependent abundance indices used in this assessment were uncertain in part due to the lack of an effective sampling methodology.
- Implement a systematic age sampling program for the general recreational sector. Age samples were important in this assessment for identifying strong year classes but sample sizes were relatively small and disparate in time and space.
- Better characterize reproductive parameters including age at maturity, batch fecundity, spawning seasonality, and spawning frequency.
- Better characterize the genetic structure of the stock and evaluate the possibility of local population structure.
- Better characterize the migratory dynamics of the stock and the degree of fidelity to spawning areas.
- Age-dependent natural mortality was estimated by indirect methods for this assessment of cobia. Tag-recapture programs for cobia exist and may prove useful for estimating mortality.
- Obtain MRIP intercept numbers at the DW for cobia and other rarely caught species.

Review Workshop

- Motives and selectivity of discarding fish by fishers. The current data compilation exercises appear to concentrate on estimating discard mortality, without any consideration of the selective impact of discarding. It would be beneficial to broaden our understanding on the

motives for discarding and the selectivity imposed by the behavior to aid considerations of size at age and what appropriate assumptions could be included in the assessment model.

- Further analysis of the interactions of length/age and maturity of Cobia. The number of observations that drive the maturity ogive is very low, even relative to the total number of Cobia aged. The minimum landing length appears to impact on the collection of potential samples and is above the likely length of 50% mature. A research approach needs to be developed that strengthens the estimation of the maturity ogive by considering the interaction of size and age and the impact of variability in female maturity on the estimation of benchmarks/reference points. This research will probably have to increase the number of observations of maturity status of 1, 2, 3 and 4 year old fish by sex.
- The DW recommended tagging to study movement patterns. The RW suggests that a tagging program may also help to inform the cobia stock assessment. The fishery and biology of cobia seems to be conducive for a successful tagging program. The fishery for cobia is currently dominated by a recreational fishery with a two-fish bag limit and a minimum landing size, resulting in a large portion of discarded catch. Discarded cobia appear to have high survival (e.g., 95% discard survival assumed in the assessment). Therefore, a tagging program conducted as an industry partnership could release tagged fish from normal fishing operations. Few cobia are discarded per trip, so the additional costs and resources required per trip would be expected to be small, and the data recording aspects at sea would be minimal. The impact on the fishing operations would be anticipated to be negligible. The major costs would be organization, tags, data collation, outreach, a reporting system for recaptured tags, and subsequent data analysis. Industry participation rates might be high if information is provided back to participants, and their collaboration improves stock assessment and fishery management. This information should improve estimates of discard numbers and potentially fish sizes. Estimates of discard mortality may be possible from initial Z from early returns compared with Z on later returns, though this will be compounded with selection. Estimates of Z or tag recovery rate on older ages will help to inform on the appropriate selection function to be used in the assessments could be obtained from ratio of tag returns on from one year to the next. Using tag return data the total mortality $Z(i,j,y)$ between year i and year j, of fish belonging to year class y is obtained using the Jolly-Seber estimator (see Ricker, 1975):

$$Z(i,j,y) = \log\left\{\frac{r(i,k,y)}{r(j,k,y)} \cdot \frac{R(j,y)}{R(i,y)}\right\} (1)$$

where $R(i,y)$ is the number of tagged fish of year class y that were released in year i, $R(j,y)$ is the number of tagged fish of the same year class that were released in year j ($j > i$) and $r(j,k,y)$ is the numbers of such tagged fish that were recaptured in the years k summed over all $k > j$. Though variability may be caused by variation in initial tagging losses, small numbers of recovered tags and errors in ageing (Antsalo, 2006). If resources are available consideration should be given to coupling two types of tagging: 1) high volume, low cost tagging would be most informative for estimates of Z that would help with population level estimates of total mortality and possibly selection and natural mortality; 2) high cost, electronic tagging might give more detail on migration. Of the two methods, the high volume approaches are more likely to be informative for management parameters at a population level.

References:

- Antsalo, M. 2006. Abundance estimation of the Northeast Atlantic mackerel (*Scomberscombrus*) with use of Norwegian tag data. University of Bergen, Department of Biology Bergen Norway. Dissertation 64 pp.

Ricker, W.E., 1975. Computation and interpretation of biological statistics of fish populations. Bull. Fish. Res. Board Can. 191: 1–382.

South Atlantic Spanish Mackerel:

Data Workshop

Life History

- Collect Spanish mackerel maturity data from both regions and both sexes from specimens approximately 275 mm FL and lower to be staged via histological methods.

Commercial Statistics

Although under the category of research recommendations, this list is not research per se, but rather suggestions to improve data collection. The first three recommendations were taken verbatim from the SEDAR17 DW report.

- Need observer coverage for the fisheries for Spanish mackerel (gillnets, castnets (FL), handlines, poundnets, and shrimp trawls for bycatch):
 - 5-10% allocated by strata within states
 - possible to use exemption to bring in everything with no sale
 - get maximum information from fish
- Expand TIP sampling to better cover all statistical strata
 - Predominantly from Florida and by gillnet & castnet gears
 - In that sense, we have decent coverage for lengths
- Trade off with lengths versus ages, need for more ages (i.e., hard parts)
- Consider the use of VMS to improve spatial resolution of data
- During discussions at the data workshop it was noted that the logbook categories for discards (all dead, majority dead, majority alive, all alive) are not useful for informing discard mortality. Consider simplified logbook language in regard to discards (e.g., list them as dead or alive)
- Uniformity between state and federal reporting systems/forms would vastly improve the ease and efficiency of data compilation.
- Establish online reporting and use logbooks as a backup.
- Establish a mechanism for identifying age samples that were collected by length or market categories, so as to better address any potential bias in age compositions.
- Compiling commercial data is surprisingly complex. As this is the 28th SEDAR, one might expect that many of the complications would have been resolved by now through better coordination among NMFS, ACCSP, and the states. Increased attention should be given toward the goal of "one-stop shopping" for commercial data.

Recreational Statistics

- Increase proportion of fish with biological data within MRFSS sampling.
- Continue to develop methods to collect a higher degree of information on released fish (length, condition, etc.) in the recreational fishery.
- Require mandatory reporting for all charter boats state and federal.
- Continue development of electronic mandatory reporting for for-hire sector.
- Continued research efforts to incorporate/require logbook reporting from recreational anglers.

- Establish a review panel to evaluate methods for reconstructing historical landings (SWAS, FWS, etc.).
- Quantify historical fishing photos for use in reconstructing recreational historical landings.
- Narrow down the sampling universe. Identify angler preference and effort. Require a reef fish stamp for anglers targeting reef fish, pelagic stamp for migratory species, and deepwater complex stamp for deep-water species. The program would be similar to the federal duck stamp required of hunters. This would allow the managers to identify what anglers were fishing for.
- Continue and expand fishery dependent at-sea-observer surveys to collect discard information, which would provide for a more accurate index of abundance.

Indices

- Collect and analyze fishery independent data for adult Spanish mackerel.
- Using simulation analysis, evaluate the utility of including interaction terms in the development of a standardized index and identify the potential effects these interaction terms have on stock assessments.

Assessment Workshop

The research recommendations from the AW panel were as follows:

- Establish a fishery-independent survey meant to capture the population trends of coastal pelagics in the south Atlantic.
- Examine how schooling or migratory dynamics may influence the catchability of the species. In particular, research the assumption of the hyperstability of indices that sample the schooling portion of the stock.
- Determine whether it is important to model both sexes in the population for assessment purposes.

Review Workshop

- Stock structure. Following on from the comments in section 2.3 of SEDAR 28, South Atlantic Spanish mackerel Section II, the review recommends that recently developed genetic techniques be utilized to investigate the stock structure of Spanish mackerel. The studies cited are relatively old, and use techniques that could be now considered antiquated and may not have the power to distinguish population structure in highly migratory species. Microsatellite information should be explored to consider both stock identity and internal population structure.
- Investigation of steepness and alternative models for the stock recruit relationship. In particular evaluate if there is newer data available on steepness from other analyses of S-R for pelagic stocks with similar reproductive strategies. However, the RP was uncertain as to how much the analysis would further inform the model or management at present.

Gulf of Mexico Cobia:

Data Workshop

Life History

1. Implement a tagging study along the entire east coast of Florida and evaluate genetic samples from the same to determine more precise stock boundaries.
2. Explore the feasibility of satellite tags for Cobia movement studies.
3. Provide genetic sampling kits to interested groups to better understand the stock division line between the Gulf and Atlantic Cobia stocks. Possible collectors of genetic samples could include Charter operators, fishing clubs and state fisheries personnel.
4. Recommend developing a tagging program for inshore and offshore South Atlantic Cobia populations. The goal would be to deploy tags inshore during the spring migration and offshore during the fall and winter to get a clearer picture of fall and spring migrations and to better identify spawning areas and aggregations.
5. Conduct research on cobia release mortality.
6. To increase overall amount of data available, have port samplers do complete workups when sampling, including otolith removal for aging, length, weight, sex, genetic sampling and record a catch location.

Commercial Statistics

The WG determined the following recommendations be added to any pending recommendations issued in SEDAR 17 that have not been addressed.

- Need expanded observer coverage for the fisheries encountering cobia
 - 5-10% allocated by strata within states
 - get maximum information from fish
- Need research methods that capture cobia in large enough numbers to create a reasonable index for young (age 0) cobia
- Expand TIP sampling to better cover all statistical strata
 - Predominantly from Florida and by hand line
 - Greater emphasis on collecting unbiased samples
- Establish a mechanism for identifying age samples that were collected by length or market categories, so as to better address any potential bias in age compositions.
- Need better information on migration patterns
- Need to address issue of fish retained for bait (undersized) or used for food by crew (how to capture in landings)
- Compiling commercial data is surprisingly complex. As this is the 28th SEDAR, one might expect that many of the complications would have been resolved by now through better coordination among NMFS, ACCSP, and the states. Increased attention should be given toward the goal of "one-stop shopping" for commercial data.

Recreational Statistics

- 1) Increase proportion of fish with biological data within MRFSS sampling.
- 2) Continue to develop methods to collect a higher degree of information on released fish (length, condition, etc.) in the recreational fishery.
- 3) Require mandatory reporting for all charter boats state and federal.
- 4) Continue development of electronic mandatory reporting for for-hire sector.
- 5) Continued research efforts to incorporate/require logbook reporting from recreational anglers.
- 6) Establish a review panel to evaluate methods for reconstructing historical landings (SWAS, FWS, etc.).

- 7) Quantify historical fishing photos for use in reconstructing recreational historical landings.
- 8) Narrow down the sampling universe. Identify angler preference and effort. Require a reef fish stamp for anglers targeting reef fish, pelagic stamp for migratory species, and deepwater complex stamp for deep-water species. The program would be similar to the federal duck stamp required of hunters. This would allow the managers to identify what anglers were fishing for.
- 9) Continue and expand fishery dependent at-sea-observer surveys to collect discard information, which would provide for a more accurate index of abundance.

Indices

None provided.

Assessment Workshop

Gulf of Mexico cobia suffers some of the same problems that make assessments of data poor species so difficult. There is not a large targeted fishery for cobia and they tend to occur mostly as an opportunistic catch. For this reason, many sources of data lacked sufficient sample sizes to be included in the assessment. Given the low frequency of positive catches pre trip, both of the fishery dependent indices of abundance and the annual estimates of recreational discards were sensitive to individual positive catches.

The majority of the length composition data, all of the age-composition data, and both indices of abundance came from the recreational fishery which is the primary fishery. The landings data are dominated by the recreational fishery; however, catches prior to 1981 are likely highly uncertain. Uncertainty in the hindcast estimates of recreational landings was not incorporated into the model and should be evaluated in future assessments.

Data on the size of discarded fish was lacking for the recreational fishery. The reef fish observer program provided some information on the size composition of released fish for the commercial fishery in recent years. This information helped in estimating the selectivity and retention parameters of the commercial fishery. Length composition data of discarded fish for the recreational fishery would have improved the assessment model.

Lack of age composition data restricted the assessment from being able to track cohorts through time or identify strong year classes. A systematic age sampling program for the recreational fishing sector would improve future assessments.

The parameters describing early growth of cobia and the selectivity pattern of the shrimp fishery had the greatest uncertainty and required extensive model diagnostics to reconcile. Additional information on the size selectivity patterns for the shrimp fishery would have improved the assessment model.

Gulf of Mexico Spanish Mackerel:

Data Workshop

Life History

None provided.

Commercial Statistics

The WG determined the following recommendations be added to any pending recommendations issued in SEDAR 17 that have not been addressed.

- Need expanded observer coverage for the fisheries for Spanish mackerel.
 - 5-10% allocated by strata within states
 - get maximum information from fish
- Need research methods that capture Spanish mackerel in large enough numbers to create a reasonable index for young (age 0) Spanish mackerel.
- Expand TIP sampling to better cover all statistical strata.
 - Predominantly from Florida and by gillnet
 - Greater emphasis on collecting unbiased samples
- Establish a mechanism for identifying age samples that were collected by length or market categories, so as to better address any potential bias in age compositions.
- Need better information on migration patterns.
- Need to address issue of fish retained for bait (undersized) or used for food by crew (how to capture in landings).
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- 9) Continue and expand fishery dependent at-sea-observer surveys to collect discard information, which would provide for a more accurate index of abundance.

Indices

None provided.

Assessment Workshop

Gulf of Mexico Spanish mackerel has a lengthy history of exploitation dating to the early late 1800s. Directed commercial gillnet fisheries have operated on this resource for well over a

hundred years and recreational fisheries more than 65 years. However detailed catch statistics on size and individual weight of removals only exists for the recent time period, since the mid to late 1980's. In addition, management measures including size limits (30.5 cm FL beginning 1983) and quotas (beginning in 1987) have resulted in discards for both fisheries.

Gulf of Mexico Spanish mackerel are not a directed target of the commercial line gear fisheries (COM_RR fleet) therefore extensive samples for length and/or age-length key characterizations are not available. Efforts should be made to obtain samples from this fleet in order to better inform future stock assessment evaluations as relates length composition and discard levels. In particular, a review of the sampling protocols for length and age – length collections is needed to better characterize the catch length and age at length compositions. In addition, attention is needed to evaluate optimal spatial sampling factors in relation to overall removals throughout the year and region.

The magnitude of discards from the recreational fleet is high and very variable over the time series for which estimates exist from the MRFSS/MRIP survey (1981 forward). Hind casting was used to develop estimates of recreational removals and discards prior to 1981 however information on uncertainty in the hind casting was not incorporated into the stock assessment. Future assessments should consider uncertainty around hind casted data.

The indices of abundance are generally flat but variable yielding little information with which to characterize abundance. In addition the additional observations of length and conditional age at length are more recent thus providing only limited history of data with which to estimate the spawner- recruit relationship during the early part of the time period. The quantity and quality of length and age composition information directly impacts the ability to estimate recruitment.

There was difficulties with estimating steepness thus the AW felt that providing benchmarks at several levels and making projections using several levels of steepness was needed.